LEZIONE 5

... CONTINUA: DERIVATA di $\mathcal{L}_2$ su SLP

$$\frac{\partial \mathcal{L}}{\partial w_{ij}} = \dots \quad \frac{\sigma(s_j) - y_j}{\|\sigma(s) - y\|} \cdot \sigma'(s_j) \cdot x_i$$

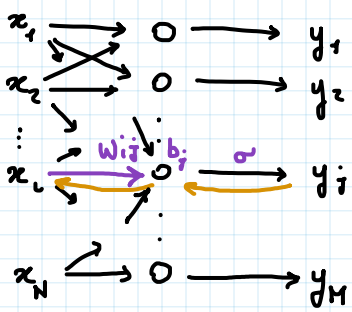
Ultimo step, derivata di $(\cdot)^2$.

$$\frac{\partial \|\sigma(s) - y\|^2}{\partial w_{ij}} = 2 \cancel{\|\sigma(s) - y\|} \cdot \frac{\sigma(s_j) - y_j}{\cancel{\|\sigma(s) - y\|}} \cdot \sigma'(s_j) \cdot x_i$$

DATASET $|D| = m$

$$\frac{\partial \mathcal{L}}{\partial w_{ij}} = \frac{1}{\cancel{2}m} \sum_{k=1}^{m} \cancel{2} \cdot \boxed{(\sigma(s_j) - y_j^{(k)}) \cdot \sigma'(s_j)} \; x_i^{(k)}$$

ERRORE sul NEURONE j di OUTPUT



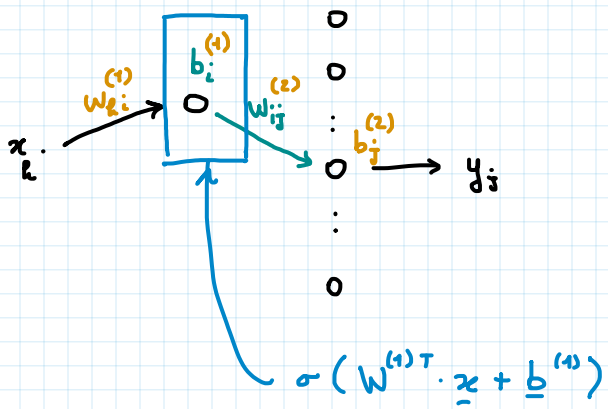
RISPETTO A y_j^* ? (Nel dataset)

$$(y_j - y_j^*) \cdot \sigma' \cdot x_i$$

$$y_j = \sigma(w_{ij} x_i + \dots + b_j)$$

$$\frac{\partial \mathcal{L}}{\partial b_j} = \frac{1}{m} \sum_{k=1}^{m} (\sigma(s_j) - y_j^{(k)}) \cdot \sigma'(s_j)$$

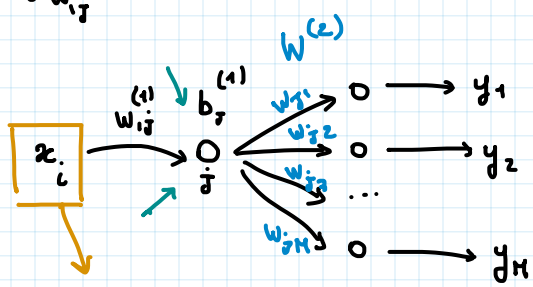
ALGORITMO di DISCESA del GRADIENTE su MLP (a 1 STRATO)



$$\frac{\partial \mathcal{L}}{\partial w_{ij}^{(2)}} = \frac{1}{m} \sum_{k=1}^{m} \left(\sigma(s_j^{(2)}) - y_j^{(k)} \right) \sigma'(s_j^{(2)}) \cdot \sigma(s_i^{(1)})$$

$$\frac{\partial \mathcal{L}}{\partial w_{ij}^{(1)}} = \,!$$

$$\frac{\partial \mathcal{L}}{\partial w_{ij}^{(1)}}$$



$$\left(w_{1j}^{(1)} x_1 + \dots + \underbrace{w_{ij}^{(1)} x_i} + \dots + w_{Nj}^{(1)} x_N + b_j^{(1)} \right)$$

$\sigma(\cdot)$

① Derivata di $\dfrac{\partial s_j^{(1)}}{\partial w_{ij}^{(1)}} = x_i$ $\qquad \dfrac{\partial s_k}{\partial w} = 0$, $k \neq j$

② Derivata di σ

$$\frac{d\sigma(s_j^{(1)})}{\partial w_{ij}^{(1)}} = \sigma'(s_j^{(1)}) \cdot x_i \qquad \frac{d\sigma(s_k)}{\partial w} = 0, \quad k \neq j$$

③ Derivata di $s_k^{(2)} = \displaystyle\sum_{v=1}^{M_1} w_{vk}^{(2)} \cdot \sigma(s_v) + b_k^{(2)}$

$$\sigma(s_j) \cdot w_{j1}^{(2)} \qquad \sigma(s_j) \cdot w_{j2}^{(2)} \qquad \dots \qquad \sigma(s_j)\, w_{jM}^{(2)}$$

$$\downarrow + b_1^{(2)} \qquad \downarrow + b_2^{(2)} \qquad \downarrow b_M^{(2)}$$

$$y_1 \qquad y_2 \qquad y_M$$

$$\frac{\partial s_k^{(2)}}{\partial w_{ij}^{(1)}} = \sum_{v=1}^{M_1} w_{vk}^{(2)} \cdot \sigma'(s_v^{(1)}) \cdot x_i = w_{jk}\, \sigma'(s_j^{(1)})\, x_i$$

$v \neq j \rightsquigarrow$ si annulla

④ Attivazione $\quad \sigma\left(s^{(2)}\right) \rightsquigarrow y$

$$\frac{d\sigma}{dw_{ij}^{(1)}} = \frac{d\sigma}{ds} \cdot \frac{\partial s}{\partial w_{ij}^{(1)}}$$

$$\frac{d\sigma\left(s_k^{(2)}\right)}{\partial w_{ij}^{(1)}} = \sigma'\left(s_k^{(2)}\right) \cdot w_{jk}^{(2)} \cdot \sigma'\left(s_j^{(1)}\right) \cdot x_i \qquad \forall k = 1, \dots, H$$

⑤ Norma euclidea:

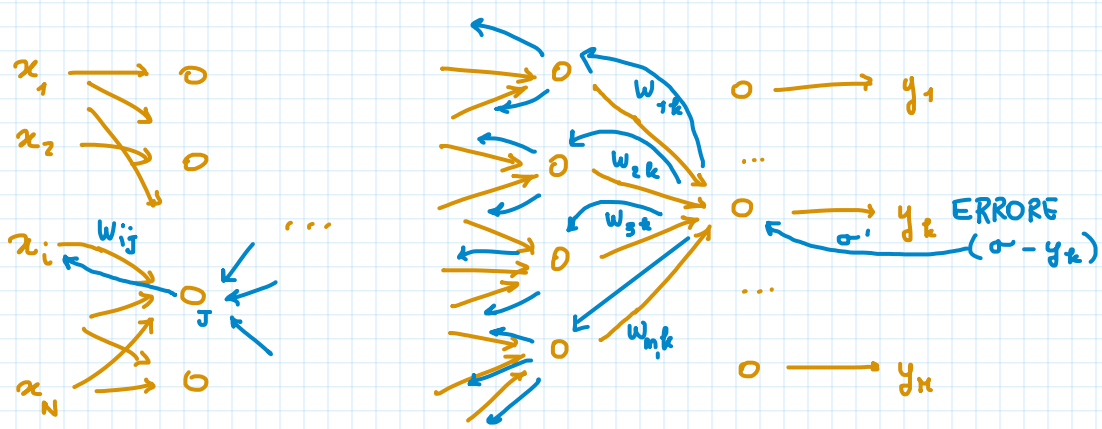
$$\frac{\partial \| \sigma(s^{(2)}) - y \|}{\partial w_{ij}^{(1)}} = \sum_{k=1}^{H} \frac{\sigma\left(s_k^{(2)}\right) - y_k}{\| \sigma(s^{(2)}) - y \|} \sigma'\left(s_k^{(2)}\right) w_{jk}^{(2)} \cdot \sigma'\left(s_j^{(1)}\right) \cdot x_i$$

FINALE:

$$\frac{\partial \mathcal{L}}{\partial w_{ij}^{(1)}} = \frac{1}{m} \sum_{k=1}^{m} \boxed{\left(\sum_{v=1}^{H} \boxed{\left(\sigma\left(s_v^{(2)}\right) - y_v^{(k)}\right) \sigma'\left(s_v^{(2)}\right)} w_{jv}^{(2)} \right) \cdot \sigma'\left(s_j^{(1)}\right) \cdot x_i^{(k)}}$$

ERRORE
sul NEURONE v di OUTPUT

ERRORE sul NEURONE interno j-ESIMO



ALGORITMO di BACK-PROPAGATION

OBIETTIVO: Calcolare in modo efficiente $\nabla_\theta \mathcal{L}$

$$\frac{\partial \mathcal{L}}{\partial W^{(1)}} \qquad \frac{\partial \mathcal{L}}{\partial \underline{b}^{(1)}}$$

$$\frac{\partial \mathcal{L}}{\partial W^{(2)}} \qquad \frac{\partial \mathcal{L}}{\partial \underline{b}^{(2)}}$$

$$\dots \qquad \dots$$

$$\frac{\partial \mathcal{L}}{\partial W^{(L)}} \qquad \frac{\partial \mathcal{L}}{\partial \underline{b}^{(L)}}$$

Def. ERRORE sullo STRATO di OUTPUT $\underline{\delta}^{(L+1)} \in \mathbb{R}^M$

$$\delta_k^{(L+1)} = \Big(\underbrace{\sigma(s_k^{(L+1)})}_{\text{CALCOLATO}} - \underbrace{\boxed{y_k}}_{\text{dal DATASET}} \Big) \cdot \sigma'(s_k^{(L+1)})$$

Def: ERRORI sugli STRATI NASCOSTI $\underline{\delta}^{(\ell)} \in \mathbb{R}^{m_\ell}$

$$\delta_j^{(\ell)} = \sum_{k=1}^{m_\ell} \delta_k^{(\ell+1)} \cdot w_{jk}^{(\ell)} \cdot \sigma'(s_j^{(\ell)})$$

EQUAZIONI della BACK-PROPAGATION

$$\boxed{\frac{\partial \mathcal{L}}{\partial w_{ij}^{(\ell)}} = \frac{1}{m} \sum_{k=1}^{m} \delta_j^{(\ell)} \cdot \underbrace{\sigma(s_i^{(\ell-1)})}}$$

Se ℓ è il I strato, $\sigma(s^{(0)}) \Rightarrow x_i$

$$\boxed{\frac{\partial \mathcal{L}}{\partial b_j^{(\ell)}} = \frac{1}{m} \sum_{k=1}^{m} \delta_j^{(\ell)}}$$

ALGORITMO di ADDESTRAMENTO di MLP

→ IPERPARAMETRI :

N_{epoche} epoche di addestramento

$\eta > 0$, $\eta \in \mathbb{R}$ coeff. di apprendimento (Learning rate)

→ INPUT : $D_{train} = \{ (\underline{x}^{(1)}, \underline{y}^{(1)}), \dots, (\underline{x}^{(m)}, \underline{y}^{(m)}) \}$ $m = |D|$

$\underline{x}^{(i)} \in \mathbb{R}^N$ e $\underline{y}^{(i)} \in \mathbb{R}^M$

→ ARCHITETTURA della RETE : L numero di strati (layers)

N dim. input, $\underbrace{n_1, n_2, \dots, n_L}_{\text{dim. strati nascosti}}$, M dim. output

for $\ell = 1, 2, \dots, L, L+1$:

$\left.\begin{matrix} W_0^{(\ell)} \\ \underline{b}_0^{(\ell)} \end{matrix}\right\}$ INIZIALIZZATI in modo CASUALE θ_0

for $i = 1, \dots, N_{epoche}$:

for $\underline{x}^{(k)} \in D_{train}$:

$\underline{a}_k^{(0)} = \underline{x}^{(k)}$ INPUT ACTIVATION

for $\ell = 1, \dots, L+1$:

$\underline{a}_k^{(\ell)} = \sigma\left(W^{(\ell)T} \cdot \underline{a}_k^{(\ell-1)} + \underline{b}^{(\ell)} \right)$

PASSO FORWARD

calcolare $\underline{\delta}_k^{(L+1)}$ OUTPUT ERROR ← rispetto a $\underline{y}^{(k)}$

for $\ell = L, L-1, \dots, 1$:

calcolare $\underline{\delta}_k^{(\ell)}$

PASSO BACKWARD

for $\ell = 1, \dots, L+1$.

$$W_i^{(\ell)} = W_{i-1}^{(\ell)} - \eta \cdot \frac{1}{m} \sum_{k=1}^{m} \underline{a}_k^{(\ell-1)} \underline{\delta}_k^{(\ell)}$$

$$\underline{b}_i^{(\ell)} = \underline{b}_{i-1}^{(\ell)} - \eta \frac{1}{m} \sum_{k=1}^{m} \underline{\delta}_k^{(\ell)}$$

AGGIORNAMENTO dei PARAMETRI θ_i

(STEP di DISC. GRAD.)

TEOREMA di APPROSSIMAZIONE UNIVERSALE

Sia $\sigma : \mathbb{R} \to \mathbb{R}$, $\sigma \in \mathcal{C}^1(\mathbb{R})$ e $\sigma \nearrow$ FUNZIONE di ATTIVAZIONE

Sia $f : K \subseteq \mathbb{R}^N \to \mathbb{R}^M$ dove K è chiuso e limitato (COMPATTO)

$\boxed{f \in \mathcal{C}^0(K)}$ FUNZIONE OBIETTIVO

Fissato $\varepsilon > 0$

Allora <u>ESISTONO</u>:

- $m_1 \in \mathbb{N}$ # neuroni I strato
- $W^{(1)} \in \mathbb{R}^{N \times m_1}$ e $W^{(2)} \in \mathbb{R}^{m_1 \times M}$] MATRICI di PESI
- $\underline{b}^{(1)} \in \mathbb{R}^{m_1}$] VETTORE di BIAS del I strato

t.c. la funzione $F : \mathbb{R}^N \to \mathbb{R}^M$

$$\boxed{F(\underline{x}) := W^{(2)T} \cdot \sigma\left(W^{(1)T} \cdot \underline{x} + \underline{b}^{(1)}\right)}$$

semplificato (MLP con 1 strato

soddisfa

$$\boxed{\max_{\underline{x} \in K} \left\| F(\underline{x}) - f(\underline{x}) \right\| < \varepsilon}$$

NON è un teorema COSTRUTTIVO $\to$ $\exists$ parametri MA NON so quali sono!